

COMPASS: Conversation Onboarding Mechanism for Project Alignment and Setup Support

Anthony Wise¹, Reese Fairchild^{1*}, Shruti Badrish^{1*}, Yashika Kachana^{1*}, Ella Cao^{1*}, Anind Dey², and Leilani Battle¹



¹Paul G. Allen School of Computer Science, University of Washington, Seattle

²The Information School (iSchool), University of Washington, Seattle

*Denotes that author is presenting this work at the symposium

Abstract

To meaningfully contribute to a research team, an individual must learn the information known by, and the skills possessed by, the team, as well as specialized knowledge for their particular scientific sub-field(s). **In this work, we investigate how AI assistance can support and accelerate the onboarding process for newcomers joining an existing research team.** We introduce a collaborative research tool called **COMPASS** that supports the onboarding of new members during routine research collaboration activities. Through a user study, we show that COMPASS enabled users to **identify knowledge gaps and explore remedies with relative ease**, demonstrating the potential for COMPASS to reduce the time and effort required for onboarding new members of a research team.

Design Goals

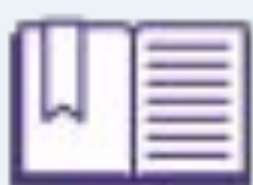
As the goals of COMPASS are to facilitate **sharing of knowledge between new and existing members of research teams**, we adapt our design goals from existing research in scaffolding science inquiry for learners [4]. Our design goals are as follows:



DG1: “Use **representations** and language that **bridge learner’s understanding**” [4]



DG3: “Use representations that **learners can inspect in different ways** to reveal important properties of underlying data” [4]

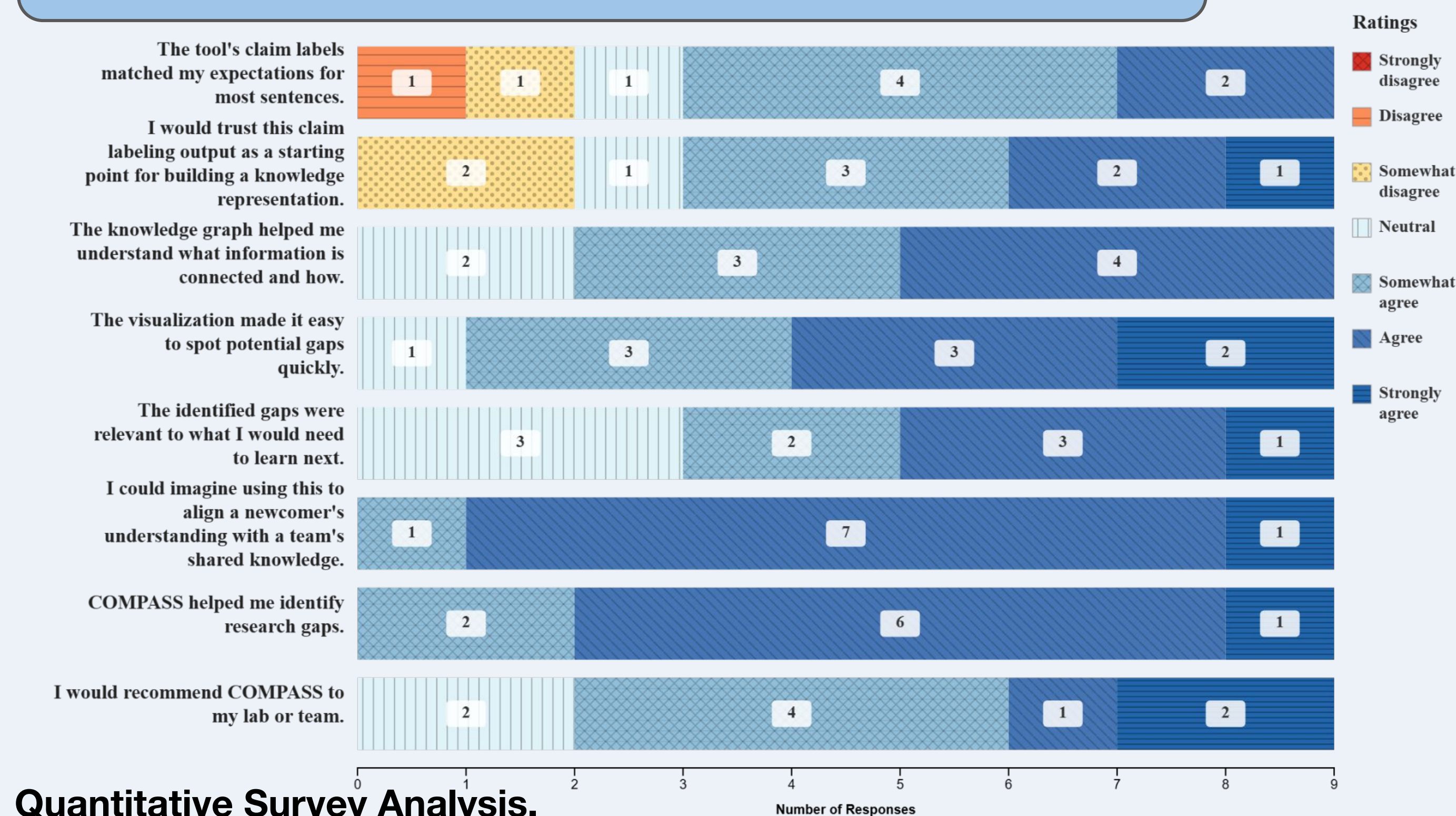


DG2: “Organize tools and artifacts around the **semantics of the discipline**” [4]



DG4: Develop the tool to **generalize to different team structures and scientific disciplines**

Evaluation

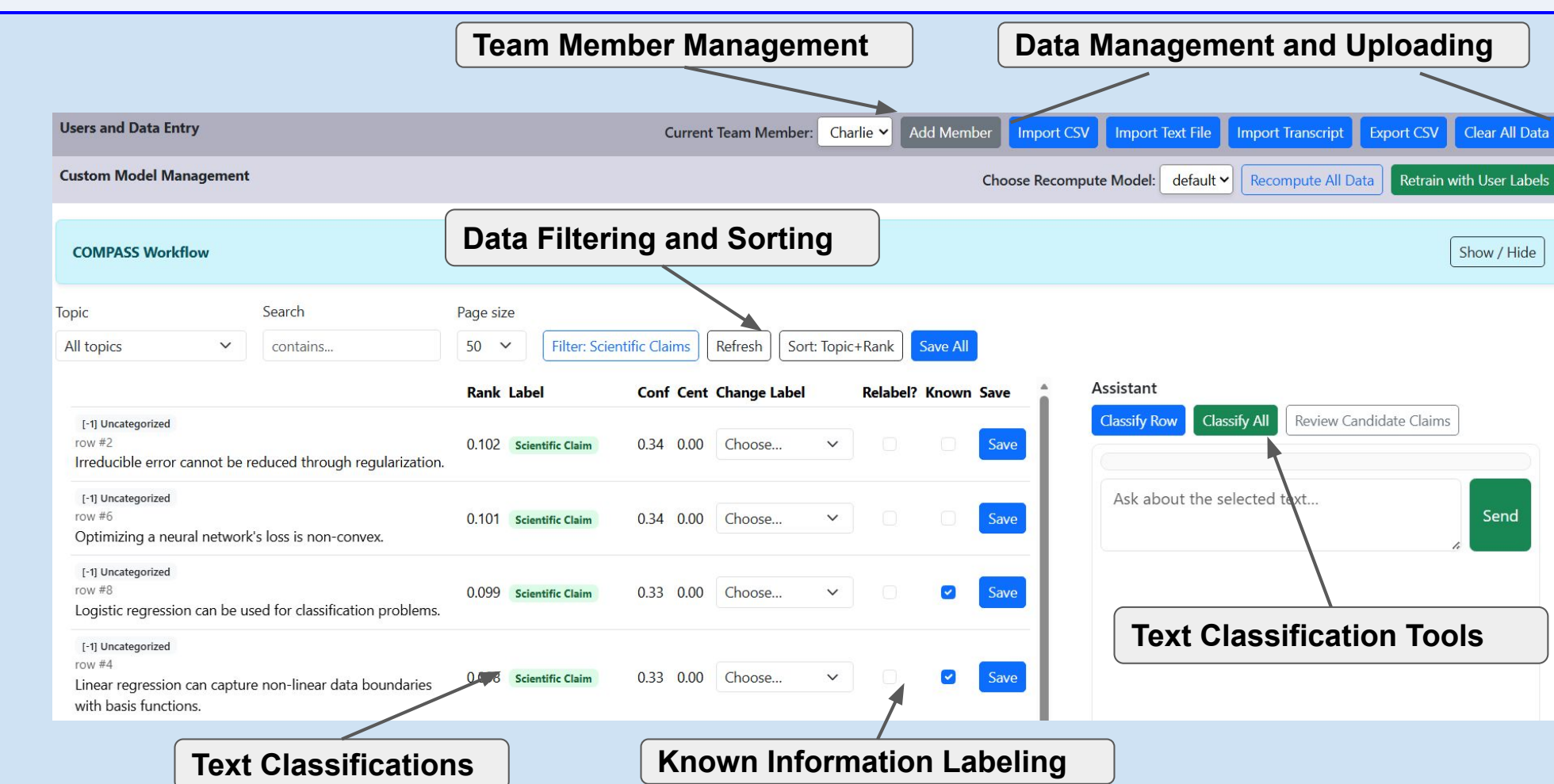


The COMPASS Tool

COMPASS helps onboard users through three steps: **detecting knowledge, identifying knowledge gaps, and filling gaps.**

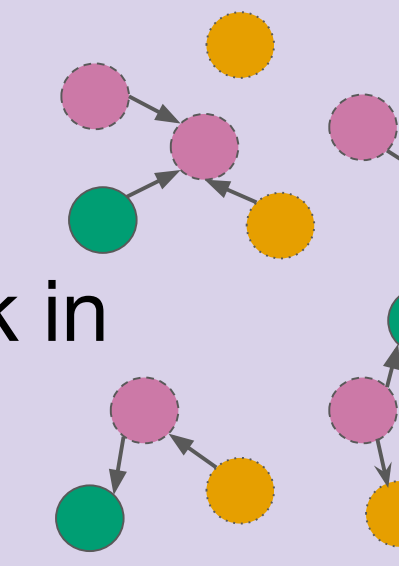
To **detect knowledge**, COMPASS:

1. Parses provided lab documents (.txt, .csv, or Zoom transcripts)
2. Automatically classifies found text as a “Scientific Claim”, “Not a Scientific Claim”, “Grey Area Claim”, or Invalid
3. Allows users to mark information they know/possess



To **identify gaps**, COMPASS:

1. Generates an interactive, modifiable knowledge graph representation of a team knowledge, inspired by past work in scholarly synthesis [1,2]
2. Selects and displays important atomic concepts that the user does not possess, along with a priority learning ranking

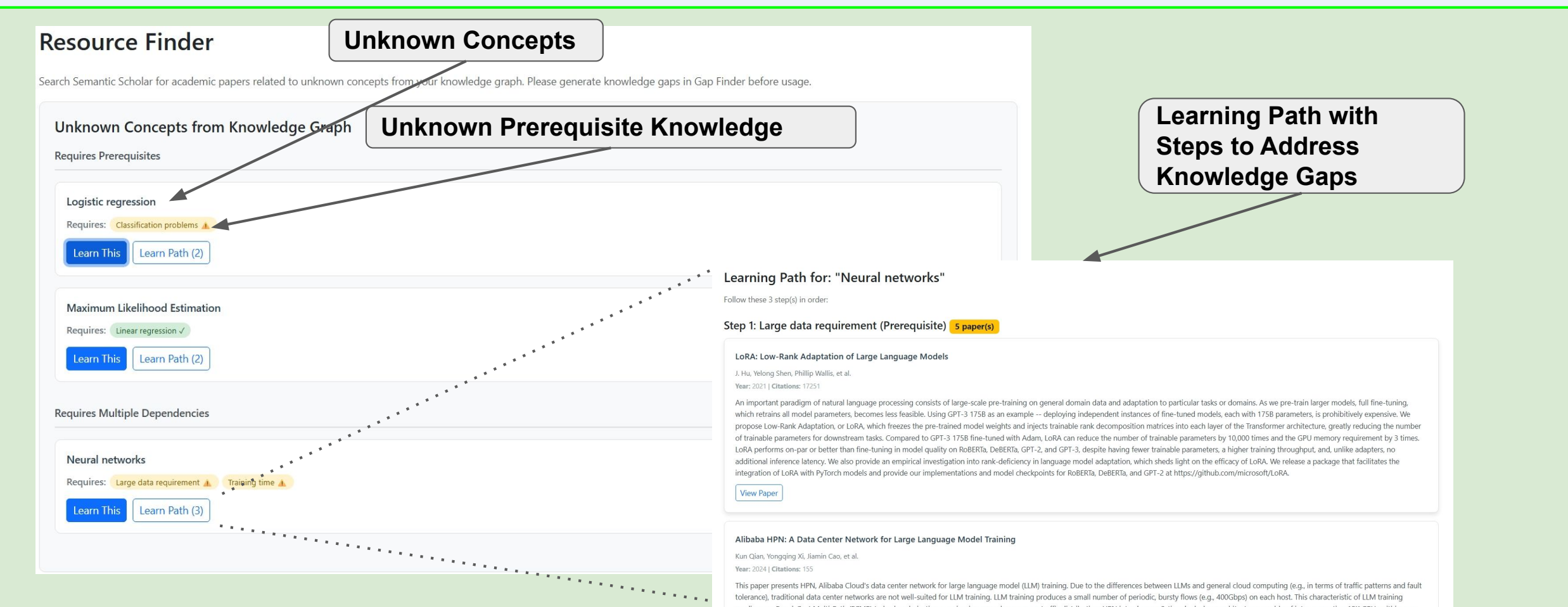


User: Charlie			
Identified Knowledge Gaps		Priority Labels	Team Members Who Possess Knowledge
#	Idea	Priority	Known by
1	L2 Norm regularizer	High	Bob
2	Neural network and its components	High	-
3	Irreducible error	High	Bob, Anna
4	Model variance and Bias	Medium	-
5	Regularization	Medium	-
6	Basis functions	Medium	-

Learning Priority Ranking

To **fill gaps**, COMPASS:

1. Recommends related scientific papers (alongside their abstracts, years, authors, and citation counts)
2. Shows which team members possess the needed knowledge, allowing users to easily contact them



To evaluate COMPASS, we conduct a **user study** with **9** researcher participants. We evaluate the tool in both **individual** and **team** settings.

Task Load Index [3]. Participants rated the burden to use the tool generally favorably, with median scores across all 7 workload dimensions falling **at or below a score of 3.0 on a 7.0 scale.**

Qualitative Interview Analysis. Participants generally liked COMPASS and noted the **novelty of the knowledge graph builder**. Issues related to claim **labeling accuracy**, recommended **papers’ quality**, and the **burden of manually labeling known claims**.

References

- [1] Chu Chan, Aakanksha Naik, Matthew Akamatsu, et al. 2024. Overview of the Context24 Shared Task on Contextualizing Scientific Claims. 12–21. doi:10.18653/v1/2024.sdp-1.3
- [2] Joel Chan, Matthew Akamatsu, David Vargas, et al. 2024. Steps Towards an Infrastructure for Scholarly Synthesis. arXiv:2407.20666
- [3] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In Human Mental Workload, Peter A. Hancock and Najmedin Meshkati (Eds.). North-Holland, Amsterdam, 139–183.
- [4] Chris Quintana, Brian J Reiser, Elizabeth A Davis, Krajcik, et al. 2018. A scaffolding design framework for software to support science inquiry. In Scaffolding. Psychology Press, 337–386.